Number of k-Combinations

- **Theorem**
  Let n and k be integers with $0 \leq k \leq n$. The number of k-combinations of a set with n distinct elements, denoted by $C(n,k)$ or $\binom{n}{k}$, is equal to

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!\,(n-k)!} = \frac{n(n-1)\ldots(n-k+1)}{k!}$$

Comment: $C(n,0) = \binom{n}{0} = \frac{n!}{0!(n-0)!} = \frac{n!}{n!} = 1$